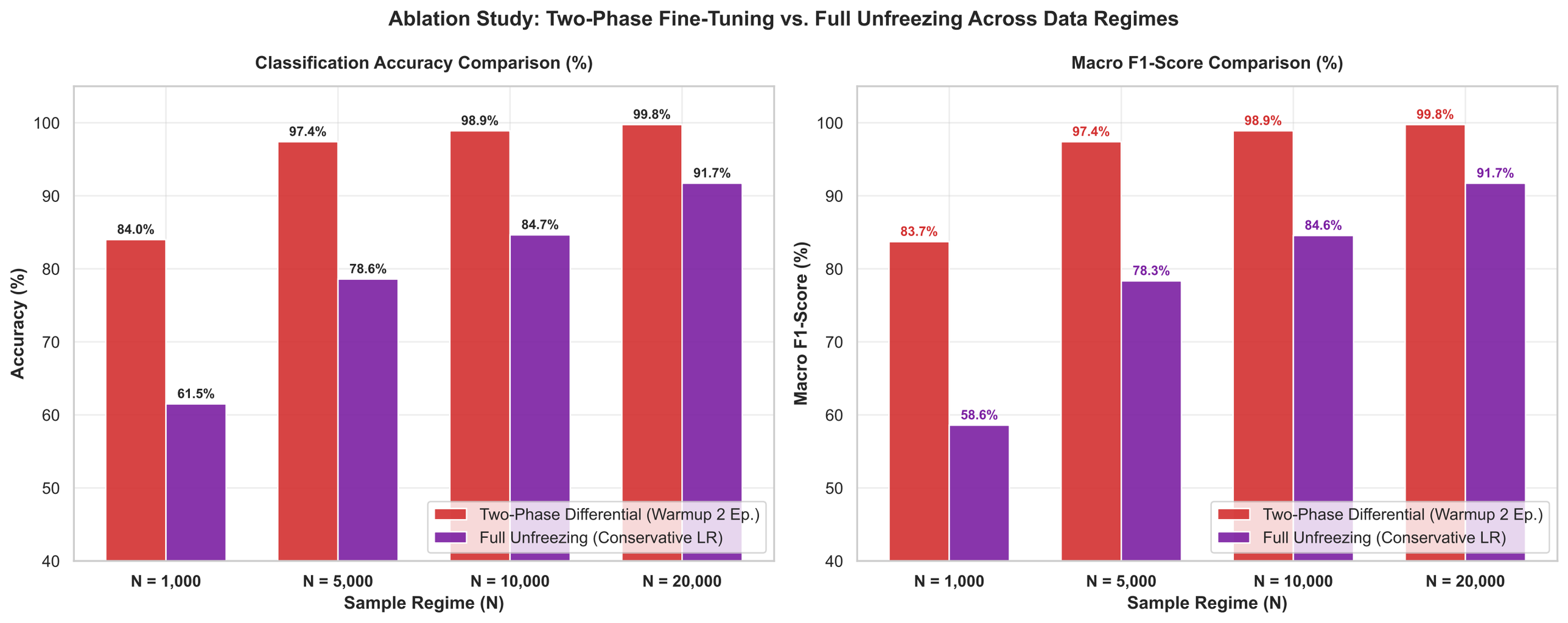
Ablation Study: Two-Phase Fine-Tuning vs. Full Unfreezing Across Data Regimes
Classification Accuracy Comparison (%)
Accuracy (%)
100
90
80
70
60
50
40
84.0%
61.5%
97.4%
78.6%
98.9%
84.7%
99.8%
91.7%
Two-Phase Differential (Warmup 2 Ep.)
Full Unfreezing (Conservative LR)
N = 1,000
N = 5,000
N = 10,000
N = 20,000
Sample Regime (N)
Macro F1-Score Comparison (%)
Macro F1-Score (%)
100
90
80
70
60
50
40
83.7%
58.6%
97.4%
78.3%
98.9%
84.6%
99.8%
91.7%
Two-Phase Differential (Warmup 2 Ep.)
Full Unfreezing (Conservative LR)
N = 1,000
N = 5,000
N = 10,000
N = 20,000
Sample Regime (N)